Lesson 2: Drawing Shapes 🎨

1. Complete code

Create a file named `lesson2.py` and copy this code.

Run it using:

python lesson2.py

2. Understand each shape

A. Rectangle

pygame.draw.rect(screen, RED, (50, 50, 150, 100))

The values mean x-position, y-position, width, and height.

B. Circle

`pygame.draw.circle(screen, BLUE, (350, 100), 50)`

The values mean center x, center y, and radius.

C. Line

`pygame.draw.line(screen, GREEN, (500, 50), (700, 150), 5)`

The coordinates specify the start and end points. The last value is thickness.

D. Ellipse

`pygame.draw.ellipse(screen, YELLOW, (50, 250, 180, 100))`

An ellipse is an oval shape drawn inside a rectangle.

lesson2.py > ...

```python

import pygame

pygame.init()

# Create the game window
screen = pygame.display.set_mode((800, 600))
pygame.display.set_caption("Lesson 2 - Drawing Shapes")

# Define colors using RGB
WHITE = (255, 255, 255)
RED = (255, 0, 0)
GREEN = (0, 200, 0)
BLUE = (0, 100, 255)
YELLOW = (255, 220, 0)
BLACK = (0, 0, 0)

running = True

while running:

    # Check events
    for event in pygame.event.get():
        if event.type == pygame.QUIT:
            running = False

    # Background
    screen.fill(WHITE)

    # 1. Draw a rectangle
    pygame.draw.rect(screen, RED, (50, 50, 150, 100))

    # 2. Draw a circle
    pygame.draw.circle(screen, BLUE, (350, 100), 50)

    # 3. Draw a line
    pygame.draw.line(screen, GREEN, (500, 50), (700, 150), 5)

    # 4. Draw an ellipse
    pygame.draw.ellipse(screen, YELLOW, (50, 250, 180, 100))

    # 5. Draw a polygon (triangle)
    pygame.draw.polygon(
        screen, GREEN,
        [(350, 250), (300, 350), (400, 350)]
    )

    # 6. Draw a rectangle outline
    pygame.draw.rect(screen, BLACK, (500, 250, 150, 100), 5)

    # Update display
    pygame.display.flip()

pygame.quit()

```

E. Polygon

```
pygame.draw.polygon(screen, GREEN, [(350,250), (300,350), (400,350)])
```

A polygon connects points to create shapes such as triangles.

3. Understand screen coordinates

Remember this rule: (0, 0) is the top-left corner of the window.

In an 800 × 600 window, x increases toward the right and y increases toward the bottom.

For example:

```
pygame.draw.circle(screen, RED, (400, 300), 50)
```

This draws a red circle at the center of an 800 × 600 window.

4. Practical challenge: Draw a car

Now let's use shapes to make a simple car!

Add this code inside your game loop, after `screen.fill(WHITE)` and before `pygame.display.flip()`. Run the program. You've drawn a car using basic shapes.

```
    # Check events
    for event in pygame.event.get():
        if event.type == pygame.QUIT:
            running = False

    # Background
    screen.fill(WHITE)

    # Car body
    pygame.draw.rect(screen, BLUE, (300, 300, 160, 60))

    # Car roof
    pygame.draw.rect(screen, BLUE, (330, 270, 100, 40))

    # Windows
    pygame.draw.rect(screen, WHITE, (340, 278, 35, 25))
    pygame.draw.rect(screen, WHITE, (383, 278, 35, 25))

    # Wheels
    pygame.draw.circle(screen, BLACK, (330, 365), 20)
    pygame.draw.circle(screen, BLACK, (430, 365), 20)

    # Update display
    pygame.display.flip()
```

